

Project 1: The AI House Digital Ecosystem (General Model)

Main Focus: Create new **Workshops** and **AI Representative** profiles:

- A. AI Representative Profiles related to workshops:** This feature manages the network of faculty members leading the AI transition within their departments.
 - **Identity:** Full name, academic title, and a professional photo.
 - **Department Link:** The specific department they represent at the University of Blida 1.
 - **AI Focus:** Their specific technical interest (e.g., Python for Research, Data Science, or Automation).
 - **Validation Status:** A field to indicate if they have completed the "Train the Trainer" session and are ready to mentor others.
- B. New Workshops (The "Training Engine"):** This allows the AI House to grow its catalog of activities dynamically.
 - **Workshop Title:** The name of the session (e.g., "Intro to Python for Biologists").
 - **Target Audience:** Specifies if the session is for department representatives, students, or the general public.
 - **Schedule & Location:** Date, time, and whether it is a physical workshop or a webinar.
 - **Resources:** A link to the JSON data or PDF materials that participants can download after the session.

1. The Mind Map (Conception)

- **The Vision:** Introduction to the AI House's mission of spreading AI literacy across the university.
- **The Training Hub:** A centralized database of all past and future activities (Workshops, Seminars, Competitions), the corresponding members and participants.
- **The Human Network:** Profiles of members (Organizers and Instructors with the related department) and the (Participants with the related department) from various departments.
- **The Partnership Grid:** Showcasing collaborations with industry and rectorate.

2. Functional Requirements (The MERN Way)

- **Database (MongoDB):**

- **Activity Collection:** Fields for Title, Type (Workshop/Seminar/Competition), Date, Description, and Status (Past/Upcoming).
- **Department Collection:** A list of all university departments to categorize training impact.
- **User/Lecturer Collection:** To store names, departments, and training history.
- **Backend (Node/Express):**
 - **Dynamic Routes:** API endpoints like /api/activities to fetch all events or /api/activities?status=upcoming to filter them.
 - **Submission Logic:** A route to handle contact messages or registration for new workshops (the related members and participants of the different departments).
- **Frontend (React):**
 - **Reusable Components:** A single EventCard.js component that displays data differently based on its "Status."
 - **Search & Filter:** A bar allowing users to find workshops by "Department" or "Topic".

3. Non-Functional & Innovative Requirements

- **Modern UI:** Responsive design (mobile-first) so faculty can check schedules on their phones.
- **Scalability:** The system must allow an Admin to add a new workshop in 30 seconds via a form.
- **Innovation (The "Pro" Touch):**
 - **Training Heatmap:** A visual chart showing which departments are most active in AI training.
 - **Automated Resource Library:** A section where documents from any "Past Event" are automatically available for download once the event date has passed.